

Here are some conventions for relations on  $X$  (i.e. subsets of  $X \times X$ ).

- If  $R \subseteq X \times X$  then we shorthand  $(a, b) \in R$  as  $aRb$ .
- We write  $\Delta X := \{(a, a) : a \in X\}$ . Note that  $\Delta X \subseteq R$  if and only if  $R$  is a reflexive relation.
- Given  $R \subseteq X \times X$  we define  $R^{-1} := \{(a, b) : (b, a) \in R\}$ . Note that  $R^{-1} \subseteq R$  if and only if  $R$  is a symmetric relation.
- Given  $R, S \subseteq X \times X$ , we define  $R \circ S \subseteq X \times X$  where  $aR \circ Sb$  iff there exists  $c \in X$  such that  $aRc$  and  $cSb$ . Note that  $R \circ R \subseteq R$  if and only if  $R$  is a transitive relation.

Remark 1: If  $R_1, R_2, R_3 \subseteq X \times X$ , then  $(R_1 \circ R_2) \circ R_3 = R_1 \circ (R_2 \circ R_3)$ .

Proof:

If  $a(R_1 \circ R_2) \circ R_3b$  then there exists  $c$  with  $a(R_1 \circ R_2)c$  and  $cR_3b$ . Next, there exists  $d$  with  $aR_1d$  and  $dR_2c$ . But now  $dR_2 \circ R_3b$ . So,  $aR_1 \circ (R_2 \circ R_3)b$ . This proves that  $(R_1 \circ R_2) \circ R_3 \subseteq R_1 \circ (R_2 \circ R_3)$ . Mirror reasoning shows the other inclusion. ■

Remark 2: If  $R, S \subseteq X \times X$  then  $(R \circ S)^{-1} = S^{-1} \circ R^{-1}$ .

Proof:  $a(R \circ S)^{-1}b$  iff  $bR \circ Sa$ . But that happens iff there exists  $c$  such that  $bRc$  and  $cSa$ , and that happens iff there exists  $c$  such that  $aS^{-1}c$  and  $cR^{-1}b$ . That is equivalent to saying that  $aS^{-1} \circ R^{-1}b$ . ■

Remark 3: If  $S, R \subseteq X \times X$  with  $\Delta X \subseteq S$  then  $R \subseteq R \circ S$  and  $R \subseteq S \circ R$ .

Why?

If  $aRb$  then  $aSa$  implies that  $aS \circ Rb$ . Similarly  $bSb$  implies that  $aR \circ Sb$ .

Remark 4: If  $R, R', S \subseteq X \times X$  and  $R \subseteq R'$  then  $S \circ R \subseteq S \circ R'$  and  $R \circ S \subseteq R' \circ S$ .

Hopefully this is obvious.

For our purposes here, the relations on  $X$  that we will care about will be relations giving a way of describing when any two elements  $a$  and  $b$  in  $X$  are "close" to each other.

A uniform structure on a set  $X$  is a filter  $\mathcal{U}$  on  $X \times X$  satisfying that:

- $\Delta X \subseteq D$  for all  $D \in \mathcal{U}$
- $D \in \mathcal{U} \implies D^{-1} \in \mathcal{U}$
- If  $D \in \mathcal{U}$  then there exists  $D' \in \mathcal{U}$  with  $D' \circ D' \subseteq D$ .

A uniform space  $(X, \mathcal{U})$  is a set  $X$  equipped with a uniform structure  $\mathcal{U}$ .

Two trivial uniform structures are as follows:

1. The discrete uniform structure on a set  $X$  is the filter consisting of all sets containing  $\Delta X$ .

2. The indiscrete uniform structure on a set  $X$  is the filter whose only element is  $X \times X$ .

In order to get more interesting uniform structures, we consider uniform structures generated by collections of sets. Unfortunately, the third bullet point in the definition of a uniform structure makes it so that an intersection of many uniform structures may not be a uniform structure. Hence, we can't define a "smallest" uniform structure generated by some collection of sets in the same way that we can for topologies. What we can do though is consider a filterbase for a uniform structure.

We say a uniformity base on a set  $X$  is a filterbase  $\mathcal{B}$  on  $X \times X$  satisfying that:

- $\Delta X \subseteq E$  for all  $E \in \mathcal{B}$
- $E \in \mathcal{B} \implies E^{-1} \in \mathcal{B}$
- If  $E \in \mathcal{B}$  then there exists  $E' \in \mathcal{B}$  with  $E' \circ E' \subseteq E$ .

Claim: The filter  $\mathcal{U}$  generated by a uniformity base  $\mathcal{B}$  is a uniform structure.

Proof:

Let  $D \in \mathcal{U}$ . Then there exists  $E \in \mathcal{B}$  with  $E \subseteq D$ . Since  $\Delta X \subseteq E$ , we can conclude  $\Delta X \subseteq D$  as well. Also, note that  $E \subseteq D \implies E^{-1} \subseteq D^{-1}$ . And since  $E^{-1} \in \mathcal{B}$  we know  $D^{-1} \in \mathcal{U}$ . Finally, let  $E' \in \mathcal{B}$  satisfy that  $E' \circ E' \subseteq E$ . Then we also have that  $E' \in \mathcal{U}$  and  $E' \circ E' \subseteq D$ . ■

Here are two\* important uniform structures defined using uniformity bases.

1. Suppose  $(X, \rho)$  is a metric space. Then we define  $\mathcal{B} := \{D_\varepsilon : \varepsilon > 0\}$  where  $D_\varepsilon := \{(x, y) \in X \times X : \rho(x, y) < \varepsilon\}$ .